

Closing the Curvature Gap: Full Transformer Hessians

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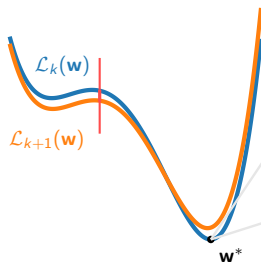
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2025

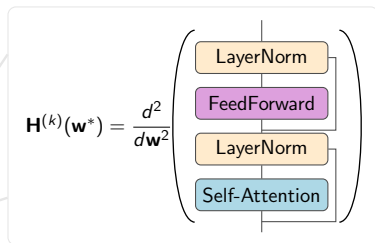
Problem: Incomplete Curvature Characterization

Why Hessians Matter

- ▶ Second-order geometry governs convergence, generalization, scaling
- ▶ Essential for understanding: optimizer behavior, critical batch size, loss landscape evolution



(a) Loss landscape convergence



(b) Hessian-based Transformer analysis

Goal: Derive explicit Hessian expressions for *all* components

Main Contributions

1. LayerNorm Calculus (Theorems 3.1–3.2)

First explicit Jacobian and Hessian for LayerNorm in vec_r notation

- ▶ Decomposition: $\text{LayerNorm}(\mathbf{X}) = \mathbf{P}(\mathbf{X})\mathbf{M}(\mathbf{X})$
- ▶ New matrix calculus lemmas for $\text{diag}^{-1}(\cdot)$, $\mathbf{A}^{\circ 1/2}$

2. Full Transformer Block Hessian (Theorem 3.5)

Complete second-order characterization via Gauss-Newton decomposition:

$$\mathbf{H}_{\text{tr}}^{(i,j)} = (\mathbf{J}_Z \otimes \mathbf{I})\xi_{ij} + (\mathbf{I} \otimes \mathbf{B}_i^\top)\mathbf{H}_Z\mathbf{B}_j$$

3. Spectral Norm Bounds & Convergence (Theorems 3.4, 3.6)

$$\|\mathbf{H}^{(k)}\|_2 \leq M \Rightarrow |\mathcal{L}_{k+1} - \mathcal{L}_k| \leq \mathcal{O}(1/k)$$

Notation Reference

Data & Dimensions

$\mathbf{X}$	Input embeddings $\in \mathbb{R}^{L \times d_V}$
L	Sequence length
d_V	Value/embedding dimension
d_K	Key/query dimension
d_{ff}	FFN hidden dimension
k	Dataset size

Parameters

$\mathbf{w}$	All parameters
$\mathbf{w}^*$	Local minimum
$\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V$	Attention weights
$\mathbf{W}_1, \mathbf{W}_2$	FFN weights

Loss & Optimization

$\mathcal{L}_k(\mathbf{w})$	Empirical loss (size k)
$\mathbf{H}^{(k)}(\mathbf{w})$	Hessian matrix
M	Spectral norm bound

Attention Components

$\mathbf{F}(\mathbf{X})$	Self-attention output
$\mathbf{A}(\mathbf{X})$	Attention matrix (softmax)
$\mathbf{G}_k$	$\frac{\partial \mathbf{F}}{\partial \mathbf{W}_k}$ (attention Jacobian)
$\Phi_{k\ell}$	$\frac{\partial \mathbf{G}_k}{\partial \mathbf{W}_\ell}$ (curvature)

Transformer Outputs

$\mathbf{Y}$	After 1st LayerNorm+residual
$\mathbf{S}$	$\text{FFN}(\mathbf{Y}) + \mathbf{Y}$
$\mathbf{Z}$	Final output (after 2nd LN)

Jacobians & Hessians

$\mathbf{J}_Z, \mathbf{H}_Z$	LN Jacobian/Hessian
$\mathbf{B}_i$	$\frac{\partial \mathbf{S}}{\partial \mathbf{W}_i}$ (1st order)
ξ_{ij}	$\frac{\partial \mathbf{B}_i}{\partial \mathbf{W}_j}$ (2nd order)

Technical

$\text{vec}_r(\cdot)$	Row-wise vectorization
$\ \cdot\ _2$	Spectral norm
$\otimes$	Kronecker product

Notation & Framework

Row-wise Vectorization (following Ormaniec et al.)

For $\mathbf{F} : \mathbb{R}^{p \times q} \rightarrow \mathbb{R}^{n \times d}$,

$$\frac{\partial \mathbf{F}}{\partial \mathbf{W}_i} := \frac{\partial \text{vec}_r(\mathbf{F})}{\partial \text{vec}_r(\mathbf{W}_i)^\top} \in \mathbb{R}^{nd \times p_i q_i}, \quad \frac{\partial^2 \mathbf{F}}{\partial \mathbf{W}_i \partial \mathbf{W}_j} \in \mathbb{R}^{(nd \cdot p_i q_i) \times p_j q_j}$$

Transformer Block

$$\begin{aligned} \mathbf{Y} &= \text{LayerNorm}(\mathbf{X} + \mathbf{F}(\mathbf{X})), \quad \mathbf{F}(\mathbf{X}) = \mathbf{A}(\mathbf{X})\mathbf{X}\mathbf{W}_V \\ \mathbf{Z} &= \text{LayerNorm}(\mathbf{Y} + \text{FFN}(\mathbf{Y})), \quad \text{FFN} = \sigma(\mathbf{Y}\mathbf{W}_1)\mathbf{W}_2 \end{aligned}$$

Gauss-Newton Decomposition

$$\frac{\partial^2(\mathcal{L} \circ f)}{\partial \mathbf{W}_i \partial \mathbf{W}_j} = \underbrace{\frac{\partial f}{\partial \mathbf{W}_i}^\top \frac{\partial^2 \mathcal{L}}{\partial f^2} \frac{\partial f}{\partial \mathbf{W}_j}}_{\text{Outer-product}} + \underbrace{\left(\frac{\partial \mathcal{L}}{\partial f} \otimes \mathbf{I} \right) \frac{\partial^2 f}{\partial \mathbf{W}_i \partial \mathbf{W}_j}}_{\text{Functional}}$$

Challenge 1: LayerNorm Jacobian

Problem

LayerNorm is *nonlinear* (row-wise normalization):

$$\text{LayerNorm}(\mathbf{X})_{i,j} = \gamma_j \frac{\mathbf{X}_{i,j} - \mu_i}{\sqrt{\sigma_i^2 + \epsilon}} + \beta_j$$

My Approach: Product Rule Decomposition

Rewrite as $\text{LayerNorm}(\mathbf{X}) = \mathbf{P}(\mathbf{X})\mathbf{M}(\mathbf{X})$ where:

- ▶ $\mathbf{M}(\mathbf{X}) = \mathbf{X} - \frac{1}{d_V} \mathbf{X} \mathbf{1} \mathbf{1}^\top$ (centering, linear)
- ▶ $\mathbf{P}(\mathbf{X}) = \text{diag}^{-1}(\sigma(\mathbf{X}))$ (scaling, nonlinear)

Theorem 3.1 (LayerNorm Jacobian)

$$\frac{\partial \text{LayerNorm}}{\partial \mathbf{X}} = (\mathbf{P} \otimes \mathbf{I}) \mathbf{G} + (\mathbf{I} \otimes \mathbf{M}^\top) \frac{\partial \mathbf{P}}{\partial \mathbf{X}}$$

where $\mathbf{G} = \mathbf{I} - \frac{1}{d_V} (\mathbf{I} \otimes \mathbf{1} \mathbf{1}^\top)$ and $\frac{\partial \mathbf{P}}{\partial \mathbf{X}}$ requires new lemmas

Challenge 2: LayerNorm Hessian

Technical Obstacles

1. $\frac{\partial \mathbf{P}}{\partial \mathbf{X}}$ involves $\text{diag}^{-1}(\sigma(\mathbf{X}))$ — need inverse and Hadamard-root derivatives
2. Nested Kronecker products from matrix-product rule
3. Chain through $\sigma(\mathbf{X}) = \frac{1}{\sqrt{d_V}}(\mathbf{M}^{\circ 2} \mathbf{1})^{\circ 1/2}$

New Matrix Calculus Lemmas (Appendix A)

- ▶ Proposition A.5: $\frac{\partial \mathbf{D}^{-1}}{\partial \mathbf{D}} = -\mathbf{D}^{-1} \otimes \mathbf{D}^{-\top}$
- ▶ Proposition A.7: $\frac{\partial \mathbf{A}^{\circ 2}}{\partial \mathbf{A}} = 2 \text{diag}(\text{vec}_r(\mathbf{A}))$
- ▶ Proposition A.8: $\frac{\partial \mathbf{A}^{\circ 1/2}}{\partial \mathbf{A}} = \frac{1}{2} \text{diag}^{-1}(\text{vec}_r^{1/2}(\mathbf{A}))$

Theorem 3.2 (LayerNorm Hessian)

$$\frac{\partial^2 \text{LayerNorm}}{\partial \mathbf{X}^2} = (\mathbf{P} \otimes \mathbf{I} \otimes \mathbf{I}) \frac{\partial^2 \mathbf{M}}{\partial \mathbf{X}^2} + (\mathbf{I} \otimes \mathbf{G}^\top) \frac{\partial (\mathbf{P} \otimes \mathbf{I})}{\partial \mathbf{X}} + \dots$$

Main Result: Full Transformer Block Hessian

Theorem 3.5 (Transformer Block Hessian)

For output $\mathbf{Z} = \text{LayerNorm}(\mathbf{S})$, $\mathbf{S} = \text{ReLU}(\mathbf{Y}\mathbf{W}_1)\mathbf{W}_2 + \mathbf{Y}$:

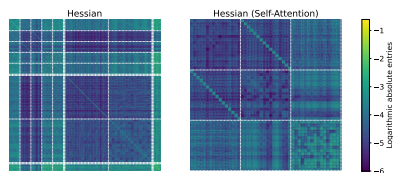
$$\mathbf{H}_{\text{tr}}^{(i,j)} = (\mathbf{J}_Z \otimes \mathbf{I}_{n_i}) \boldsymbol{\xi}_{ij} + (\mathbf{I} \otimes \mathbf{B}_i^\top) \mathbf{H}_Z \mathbf{B}_j$$

where:

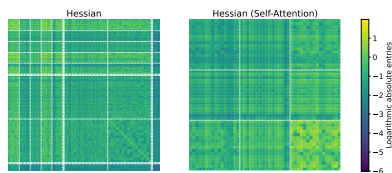
- ▶ $\mathbf{J}_Z = \frac{\partial \text{LayerNorm}(\mathbf{S})}{\partial \mathbf{S}}$ (Theorem 3.1)
- ▶ $\mathbf{H}_Z = \frac{\partial^2 \text{LayerNorm}(\mathbf{S})}{\partial \mathbf{S}^2}$ (Theorem 3.2)
- ▶ $\mathbf{B}_i = \frac{\partial \mathbf{S}}{\partial \mathbf{W}_i}$ (Theorem 3.3)
- ▶ $\boldsymbol{\xi}_{ij} = \frac{\partial \mathbf{B}_i}{\partial \mathbf{W}_j}$ (second-order interactions)

Key insight: Almost-everywhere symmetry $\mathbf{H}_{\text{tr}}^{(i,j)} = \mathbf{H}_{\text{tr}}^{(j,i)}$ because ReLU Hessian is zero a.e. (Lemma A.9), all other maps multilinear in parameters

Block-wise Structure of $\mathbf{H}_{tr}$



Initialized model



After training

Key Observations

- ▶ **Values-Values** blocks dominate in both cases
- ▶ Training amplifies heterogeneity (magnitudes increase but structure persists)
- ▶ Pure-FFN blocks show weaker curvature (ReLU linearity)

Novelty: Complete characterization beyond Ormaniec's attention-only analysis

Spectral Norm Bounds

Theorem 3.4 (Self-Attention)

$$\|\mathbf{H}_i(\mathbf{w}^*)\|_2 \leq M_{\text{SA}} = 3 \max \left(\frac{2L\|\mathbf{X}\|_2^2}{d_V}, \frac{8\|\mathbf{W}_K\|_2^2\|\mathbf{W}_V\|_2^2\|\mathbf{X}\|_2^6}{L^3d_Vd_K} + \dots \right)$$

Theorem 3.6 (Full Transformer)

Each block (i, j) satisfies:

$$\|\mathbf{H}_{\text{tr}}^{(i,j)}\|_2 \leq \|\mathbf{J}_Z\|_2 \|\boldsymbol{\xi}_{ij}\|_2 + \|\mathbf{B}_i\|_2 \|\mathbf{H}_Z\|_2 \|\mathbf{B}_j\|_2$$

Full Hessian: $\|\mathbf{H}_{\text{tr}}\|_2 \leq 5 \max_{i,j} \|\mathbf{H}_{\text{tr}}^{(i,j)}\|_2$ (5 parameter groups)

Proof technique: Submultiplicativity of spectral norm +
Kronecker identity $\|\mathbf{A} \otimes \mathbf{B}\|_2 = \|\mathbf{A}\|_2 \|\mathbf{B}\|_2$

Loss Landscape Convergence

Setup

- ▶ Empirical loss: $\mathcal{L}_k(\mathbf{w}) = \frac{1}{k} \sum_{i=1}^k l_i(\mathbf{w})$
- ▶ Taylor expansion at shared minimum $\mathbf{w}^*$ (Assumption 3.1):

$$\mathcal{L}_k(\mathbf{w}) \approx \mathcal{L}_k(\mathbf{w}^*) + \frac{1}{2}(\mathbf{w} - \mathbf{w}^*)^\top \mathbf{H}^{(k)}(\mathbf{w}^*)(\mathbf{w} - \mathbf{w}^*)$$

Theorem 3.7 (Convergence)

$$|\mathcal{L}_{k+1}(\mathbf{w}) - \mathcal{L}_k(\mathbf{w})| \leq \frac{2L}{k+1} + \frac{M\|\mathbf{w} - \mathbf{w}^*\|_2^2}{k+1}$$

where $L = \sup l_i$ and M from Theorem 3.4 or 3.6

Implication: $\mathcal{O}(1/k)$ decay rate provides sample-size guidance for practitioners

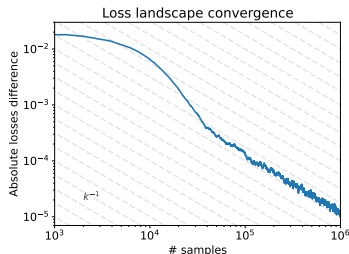
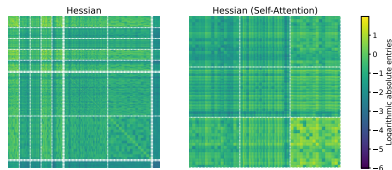
Experimental Validation

Setup

- ▶ ViT on MNIST (1 block, $d_V = 16$) and CIFAR-100 (8 blocks, $d_V = 128$)
- ▶ Train to $> 50\%$ validation accuracy
- ▶ Measure Hessian blocks via curvlinops

Observation 1: Values-Values blocks exhibit largest magnitudes (aligned with theory)

Observation 2: $|\mathcal{L}_{k+1} - \mathcal{L}_k|$ follows log-linear decay in log-log plot



Implications & Future Directions

Broader Impact

1. **Second-order optimization:** Enables curvature-aware preconditioning (e.g., per-block learning rates scaled by $\|\mathbf{H}^{(i,j)}\|_2^{-1}$)
2. **Scaling laws:** Mechanistic link between Hessian spectrum and data-model tradeoffs (compute-optimal training)
3. **Sample complexity:** Practical guidance for data-scarce domains (medical imaging, scientific discovery)

Limitations & Extensions

- ▶ Current: single-head, post-norm, MSE loss, local analysis
- ▶ Future: multi-head, pre-norm, cross-entropy, multi-layer interactions

Conclusion

Contribution: First complete second-order characterization of Transformer blocks via explicit LayerNorm/FFN Hessians

Significance: Bridges theory (matrix calculus) and practice (scaling laws, optimization)